

Wind Tunnel Experiment

Investigating the aerodynamic characteristics of the Clark Y-14 airfoil in low subsonic flow

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Abstract

This experiment investigated the aerodynamic performance of the Clark Y-14 airfoil in a low-speed, incompressible wind tunnel environment. The primary objective was to determine lift, drag, and pitching moment characteristics across a range of freestream velocities and angles of attack, including pre-stall and post-stall behavior.

Wind tunnel calibration was first performed to establish the relationship between percent fan power input and freestream velocity using Pitot-static measurements and Bernoulli's equation. The airfoil's aerodynamic characteristics were then evaluated using two independent measurement techniques. Surface pressure distributions were obtained through pressure taps distributed along the chord, allowing aerodynamic force and moment coefficients to be computed through numerical integration. In parallel, a strain-gage sting balance provided direct measurements of axial force, normal force, and pitching moment, which were uncoupled and transformed to obtain lift and drag coefficients. The wake-survey method was additionally employed to estimate drag through downstream momentum deficit analysis.

All measurements recorded by the LabVIEW acquisition system represent the mean of 1000 samples, with associated standard deviations used to quantify statistical uncertainty. Freestream velocity uncertainty was determined through analytical propagation of error applied to Bernoulli's equation, incorporating pressure and density uncertainties. Derived aerodynamic quantities, including pressure coefficients, force coefficients, and moment coefficients, were obtained using root-sum-square propagation methods based on partial derivative formulations. Reported uncertainties correspond to one standard deviation (1σ), representing a 68% confidence interval for normally distributed measurements, and later on a two standard deviation uncertainty (2σ) representing a 95% confidence for normally distributed results .

Comparison between pressure-integration, wake-survey, and direct sting-balance measurements provided internal validation of the experimental methods. The agreement between independent techniques, within quantified uncertainty bounds, confirms the reliability of the wind tunnel calibration and the consistency of the measured aerodynamic behavior with classical subsonic airfoil theory.

Introduction

Experimental aerodynamics relies on accurate, repeatable measurements of velocity, pressure, force, and moment within a controlled flow environment. Low-speed wind tunnel testing provides the means to evaluate aerodynamic performance under subsonic conditions while maintaining precise control over freestream properties. However, the accuracy of any aerodynamic coefficient derived from wind tunnel data depends fundamentally on correct determination of the freestream dynamic pressure: $q_{\infty} = \frac{1}{2}\rho_{\infty} U_{\infty}^2$ (1)

Because lift, drag, and moment coefficients are normalized by q_{∞} , even small uncertainty in velocity produces amplified error in all reported nondimensional quantities. For this reason, wind tunnel calibration is the foundational step in aerodynamic experimentation and must precede all force and pressure measurements.

The MAE 175A wind tunnel laboratory consisted of three sequential experimental phases designed to characterize the aerodynamic behavior of a Clark Y-14 airfoil under subsonic flow conditions. The experiments integrated indirect pressure-based measurements and direct force-balance measurements to evaluate lift, drag, pitching moment, and center of pressure location over a range of angles of attack extending beyond stall.

In Week 1, the wind tunnel test section was calibrated using a Pitot-static probe. The Pitot tube measures the pressure difference between stagnation and static pressure, which, under steady incompressible flow conditions, is related to freestream velocity through Bernoulli's equation (2). The velocity was determined from: $U_{\infty} = \sqrt{\frac{2\Delta P}{\rho_{\infty}}}$ (2) where $\Delta P = P_{\text{total}} - P_{\text{static}}$. The measured pressure differential was converted to velocity, and ambient air density was determined from the ideal gas law using recorded atmospheric pressure and temperature. A weighted linear calibration curve relating percent power input to freestream velocity was constructed. Because each recorded value represented the mean of 1000 samples, statistical uncertainty was quantified using standard error of the mean. Propagation of uncertainty through Bernoulli's equation and the linear regression model provided predicted velocity uncertainty for all subsequent testing conditions.

In Week 2, the pressure distribution over the Clark Y-14 airfoil was measured at multiple freestream velocities and angles of attack. Surface pressure taps provided local static pressure measurements relative to the freestream static pressure. These data were nondimensionalized to obtain the pressure coefficient distribution.

Integration of the pressure coefficient distribution along the chord yielded the normal force coefficient and moment coefficient about the leading edge. Lift and drag coefficients were determined from the normal and axial components through geometric transformation. The center of pressure was computed from the ratio of moment to lift. Additionally, the wake-survey method was employed to determine drag coefficient by evaluating the downstream momentum deficit. This approach provided an independent drag estimate based on conservation of momentum in the wake.

In Week 3, aerodynamic forces and pitching moments were measured directly using a strain-gage sting balance mounted in the wind tunnel test section. The sting balance provided axial force, normal force, and pitching moment measurements derived from amplified strain-gage bridge outputs. Because the measured quantities were not independent, empirical uncoupling relations were applied to determine corrected aerodynamic loads. Lift and drag forces were then computed and nondimensionalized to obtain C_L and C_D . These direct measurements allowed comparison with pressure-integration and wake-survey results, providing cross-validation of experimental techniques.

Throughout the three-week investigation, all derived quantities were accompanied by 1σ uncertainty bounds corresponding to a 68% confidence interval for normally distributed measurements. Error propagation was carried out analytically using partial derivative methods and root-sum-square combinations to ensure dimensional consistency and statistical rigor.

The overall objective of the three-week investigation was to characterize the aerodynamic performance of the Clark Y-14 airfoil under subsonic, incompressible conditions across a range of Reynolds numbers and angles of attack, including pre-stall and post-stall behavior. Particular emphasis was placed on consistency between independent measurement methods, rigorous uncertainty analysis, and adherence to dimensional accuracy. Through calibration, pressure integration, wake momentum analysis, and direct force measurement, the laboratory sequence provided a comprehensive experimental evaluation of lift, drag, and pitching moment characteristics in accordance with classical aerodynamic theory.

Theory

Fundamental Assumptions and Flow Model

The wind tunnel experiments were conducted under low-speed subsonic conditions. For Mach numbers less than 0.3, compressibility effects are negligible; thus, air may be treated as incompressible with constant density. The flow within the test section was therefore modeled as steady, incompressible, and inviscid outside boundary layers. Under these assumptions, the governing equations reduce to conservation of mass and Euler's equation along a streamline. For incompressible flow, $\nabla \cdot V = 0$ (3)

And integration of the inviscid momentum equation along a streamline yields Bernoulli's

Equation: $p + \frac{1}{2}\rho V^2 = \text{constant}$ (4)

This relation forms the theoretical basis for velocity measurement, dynamic pressure calculation, and aerodynamic coefficient normalization.

Wind Tunnel Calibration using a Pitot Tube

A pitot-static probe measures stagnation pressure p_t and static pressure p_s . At the stagnation point, velocity is zero. Applying Bernoulli's equation between the freestream and the stagnation point gives: $p_t = p_s + \frac{1}{2}\rho_\infty U_\infty^2$ (5)

Defining the measurable pressure difference, $\Delta P = p_t - p_s$ (6)

The freestream velocity is obtained as $U_\infty = \sqrt{\frac{2\Delta P}{\rho_\infty}}$ (2)

This equation was used to generate the calibration curve relating wind tunnel power setting to freestream velocity. The physical interpretation is direct: higher dynamic pressure corresponds to higher kinetic energy per unit volume of the flow.

Air Density From Ideal Gas Law

Since velocity depends explicitly on density, accurate density evaluation is required. Air density was determined from the ideal gas relation: $\rho_\infty = \frac{P_{atm}}{RT}$ (7)

where P_{atm} is atmospheric pressure and T is absolute temperature. An increase in temperature reduces density and therefore increases calculated velocity for the same measured pressure differential. This dependency directly affects dynamic pressure and all aerodynamic coefficients.

Dynamic Pressure and Reynolds Number

The dynamic pressure is defined as: $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2$ (1). This term represents the kinetic energy per unit volume of the freestream and is the scaling parameter used in defining nondimensional aerodynamic coefficients. The Reynolds number based on chord length is:

$Re = \frac{\rho_\infty U_\infty c}{\mu}$ (8)

Reynolds number governs boundary layer thickness, transition, separation behavior, and stall characteristics. Increasing freestream velocity increases Reynolds number, delaying separation and typically increasing maximum lift coefficient.

Pressure Coefficient

To nondimensionalize surface pressures, the pressure coefficient is defined as:

$C_p = \frac{p - p_\infty}{q_\infty}$ (9) A negative C_p indicates suction relative to the freestream. Strong suction peaks

near the leading edge correspond to high lift production. The distribution of $C_p(x)$ over the chord determines the resultant aerodynamic loading.

Force components and Coordinate Transformation

The pressure distribution produces resultant forces normal and axial to the chord:

$$C_N = \frac{N}{q_\infty c} \text{ (10a) }, C_A = \frac{A}{q_\infty c} \text{ (10b)}$$

These components are related to lift and drag through geometric transformation:

$$L = N \cos \alpha - A \sin \alpha \text{ (11a)}$$

$$D = N \sin \alpha + A \cos \alpha \text{ (11b)}$$

Lift acts perpendicular to the freestream and drag acts parallel to it. At small angles of attack, A is typically much smaller than N, and lift is dominated by force contribution.

Pressure Integration for Aerodynamic Forces

The normal force coefficient is obtained by integrating the pressure difference between the lower and upper surfaces. There were $n = 10$ ports along the upper airfoil and $m = 9$ ports along the lower airfoil that include the port at the leading edge, and because this integration was normalized by the chord length, the step size was 0.1 as each port was distanced 10% of the chord length between each other. Therefore:

$$C_N = \frac{1}{c} \int_0^c (C_{p,l} - C_{p,u}) dx = \frac{h}{2} \sum_{i=1}^{m-1} (C_{pl,i+1} + C_{pl,i}) - \frac{h}{2} \sum_{i=1}^{n-1} (C_{pu,i+1} + C_{pu,i}) \text{ (12)}$$

This integral represents the net pressure loading along the chord.

The axial force coefficient is treated as negligible.

These expressions arise from resolving pressure forces into chordwise and normal components.

Moment Coefficient and Center of Pressure

The moment about the leading edge is defined as: $C_{M,LE} = \frac{M_{LE}}{q_\infty c^2} \text{ (13a)}$

From pressure distribution,

$$C_{M,LE} = \frac{1}{c^2} \int_0^c (C_{p,u} - C_{p,l}) x dx = \frac{h}{2} \sum_{i=1}^{m-1} (C_{pl,i+1} + C_{pl,i}) - \frac{h}{2} \sum_{i=1}^{n-1} (C_{pu,i+1} + C_{pu,i}) \text{ (13b)}$$

The center of pressure location is determined from: $\frac{x_{cp}}{c} = -\frac{C_{M,LE}}{C_L} \text{ (13c)}$

As C_L approaches 0, the center of pressure becomes highly sensitive and may shift rapidly.

Wake Survey Drag Determination

Drag was independently determined using a momentum balance on the wake. For a

two-dimensional control volume, $D = \rho_\infty \int U(U_\infty - U) dy \text{ (14a)}$

This equation represents the momentum deficit in the wake. Larger velocity deficit corresponds

to higher drag.

Expressed in coefficient form: $C_D = \frac{1}{q_\infty c} \int \rho_\infty U(U_\infty - U) dy$ (14b)

This method does not rely on surface pressure integration and provides independent validation.

Sting Balance Force Uncoupling

The strain-gage sting balance outputs coupled force measurements N' , A' , and P' . Corrected forces using calibration relations:

$$N = 0.9504N' - 0.0082A' - 0.0161P'$$

$$A = 0.0608N' + 0.5912A'$$

$$P = -0.1336N' + 0.00119A' + 1.182P'$$

These relations account for structural cross-sensitivity in the measurement system.

Uncertainty Propagation

During calibration testing, each power percentage input and its respective pitot-static pressure values were recorded with their own standard deviations. Additionally, each port along the upper and lower arcs, as well as the leading edge of the chord, had their own recorded pitot-static pressure values and their standard deviations. Each port along the rake followed this exact setup. Therefore, standard deviation data collected for each power setting for calibration, velocity, and angle of attack were stored as lists. For the following uncertainty equations, any value with the subscript 'i' is denoted as an item within that list. Additionally, since the standard deviation values were recorded out of 1000 samples, they were divided by $\sqrt{1000}$ to determine a standard error of the mean.

For calibration, pitot-static pressure (ΔP) was recorded at each percentage level and converted into free-stream velocity, as denoted in equation (2). The uncertainty of the recorded free-stream velocities was calculated as follows: $\delta U_\infty = \frac{\partial U_\infty}{\partial \Delta P} * \delta \Delta P = \frac{\delta \Delta P}{\sqrt{2 \Delta P \rho_\infty}}$ (15)

A line was then fitted to the free-stream velocity data, where the slope is denoted as "a" and the y-intercept is denoted as "b". To get a predicted uncertainty for any given free-stream velocity value, this fit was considered as a weighted linear calibration. The percent-power input is denoted as P_{perc} and predicted velocity and its uncertainty are denoted as $\hat{U}_\infty$ and $\widehat{\delta U}_\infty$ respectively. Therefore, the predicted velocity is calculated as follows: $\hat{U}_\infty = aP_{perc} + b$ (16)

The following weights, weighted regression sums, and weighted residual scatter are calculated as follows.

$$\text{Weights: } w_i = \frac{1}{\delta U_{\infty,i}^2} \quad (17)$$

$$\text{Regression Sums: } S = \sum_i^n w_i \quad S_x = \sum_i^n w_i x_i \quad S_{xx} = \sum_i^n w_i x_i^2 \quad \Delta = SS_{xx} - S_x^2 \quad (18)$$

$$\text{Residual Scatter: } s_w = \sqrt{\frac{\sum_i^n w_i (U_\infty - \hat{U}_\infty)^2}{n-2}}, \quad n = 8 \quad (19)$$

The uncertainties for the “a” and “b” values as well as the covariance between “a” and “b”:

$$\delta a^2 = s_w^2 * \frac{S}{\Delta} \quad (20a), \quad \delta b^2 = s_w^2 * \frac{S_{xx}}{\Delta} \quad (20b), \quad Cov(a, b) = -s_w^2 * \frac{S_x}{\Delta} \quad (20c)$$

Therefore, the predicted free-stream velocity uncertainty, while considering residual scatter, can be reported as:

$$\widehat{\delta U}_\infty(P_{perc}) = \sqrt{P_{perc}^2 \delta a^2 + \delta b^2 + 2x_0 Cov(a, b) + a^2 \delta x_0^2 + s_w^2} \quad (21)$$

Before every reading of chordwise pressure, wake dynamic pressure, normal and axial force, and pitching moment readings, a tared reading was taken where free stream velocity was 0. Any terms that denote the tared readings will have the subscript of “o”.

$$\text{Chordwise Pressure: } \Delta P = p - p_o \quad (22a) \quad \text{Wake Dynamic Pressure: } Q = q - q_o \quad (22b)$$

$$\text{Normal/Axial Force: } N = N - N_o, \quad A = A - A_o \quad (22c)$$

$$\text{Pitching Moment: } M = M - M_o \quad (22d)$$

If any generic addition or subtraction of two terms are carried out, X and Y respectively, then the following generic expression was used to calculate their uncertainties. This expression

$$\text{was applied to the equations denoted above. } \delta(X + Y) = \sqrt{\delta X^2 + \delta Y^2} \quad (23)$$

Uncertainties for free-stream dynamic velocity and pressure coefficients were calculated as such:

$$dq_\infty = \rho_\infty U_\infty \delta U_\infty \quad \delta C_p = \frac{1}{q_\infty} \sqrt{\delta \Delta P^2 + \left(\frac{\Delta P}{q_\infty}\right)^2 * \delta q_\infty^2} \quad (24)$$

Any integration carried out during data analysis utilized the trapezoidal method, with step size $h = x_{i+1} - x_i$. Each data value within the data list was treated as a separate variable, so the partial derivative of each variable was just 1. Therefore, the formula for this type of integration and its uncertainty is as follows:

$$I = \frac{h}{2} (f(x_1) + 2 \sum_{i=2}^{n-1} f(x_i) + f(x_n)) \quad (25a) \quad \delta I = \frac{h}{2} \sqrt{(\delta f(x_1))^2 + 2 \sum_{i=2}^{n-1} (\delta f(x_i))^2 + (\delta f(x_n))^2} \quad (25b)$$

Integration was carried out to determine the normal force coefficient C_N , where the pressure coefficient is denoted as $f(x)$. Additionally, when integrating for the leading edge moment coefficient $C_{M,LE}$, $f(x)$ is denoted as the multiplication between the pressure coefficient at a given location and its distance from the leading edge. For both integrations, the step size h was 0.1.

For center of pressure x_{cp} , the uncertainty formula is as follows:

$$\delta x_{cp} = \sqrt{\left(\frac{\delta C_{M,LE}}{C_N}\right)^2 + \left(\frac{C_{M,LE}}{C_N^2}\right)^2 * \delta C_N^2} \quad (26)$$

When normalizing the wake dynamic pressure with free-stream dynamic pressure, $f(x)$ in equations 25a and 25b is denoted by the normalized wake dynamic pressure with a step size of $h = \frac{\text{rake length}}{17}$ as there are 18 equidistant ports planted lengthwise along the rake. The uncertainty of the normalized wake dynamic pressure is as follows:

$$\delta\left(\frac{q}{q_\infty}\right) = \sqrt{\left(\frac{\delta q}{q_\infty}\right)^2 + \left(\frac{q}{q_\infty^2}\right)^2 * \delta q_\infty^2} \quad (27)$$

Uncertainties of the uncoupled normal/axial forces and pitching moment are as follows:

$$\delta N = \sqrt{(0.9504 * \delta N')^2 + (0.0082 * \delta A')^2 + (0.0161 * \delta M')^2} \quad (28a)$$

$$\delta A = \sqrt{(0.0608 * \delta N')^2 + (0.05912 * \delta A')^2} \quad (28b)$$

$$\delta M = \sqrt{(0.1336 * \delta N')^2 + (0.0119 * \delta A')^2 + (1.182 * \delta M')^2} \quad (28c)$$

The uncertainties of lift and drag denoted by equations 29a and 29b are as follows. Angle of attack uncertainty was converted to radians for this calculation:

$$\delta L = \sqrt{(\cos(\alpha)\delta N)^2 + (\sin(\alpha)\delta A)^2 + ((N\cos\alpha - A\sin\alpha)\delta\alpha)^2} \quad (29a)$$

$$\delta D = \sqrt{(\sin(\alpha)\delta N)^2 + (\cos(\alpha)\delta A)^2 + ((N\sin\alpha + A\cos\alpha)\delta\alpha)^2} \quad (29b)$$

Finally, the coefficient of lift and drag was calculated as follows:

$$\delta C_L = \sqrt{\left(\frac{\delta L}{qc}\right)^2 + \left(\frac{L\delta q}{q^2 c}\right)^2 + \left(\frac{L\delta c}{qc^2}\right)^2} \quad (30a), \quad \delta C_D = \sqrt{\left(\frac{\delta D}{qc}\right)^2 + \left(\frac{D\delta q}{q^2 c}\right)^2 + \left(\frac{D\delta c}{qc^2}\right)^2} \quad (30b)$$

When collecting normal/axial force and pitching moment data, uncertainties of angle-of-attack were recorded as well as the chord uncertainty.

Experimental Procedure

Week 1: Week 1 of this experiment served to calibrate the wind tunnel to define the relationship between percent wind speed to true airspeed. To operate the wind tunnel, the rocker on the control console was used, the % SPEED was changed from 0% to 80% in increments of 5%. Data was collected at every interval using the LabView software and the Calibration VI. Using the collected pressure data, as well as the ambient pressure and temperature from the digital readings in the yellow box, airspeed was plotted versus % fan speed.

Week 2: Using the calibration relationship, the pressure distribution and wake survey systems were used to collect pressure data on the upper and lower airfoil surfaces as well as behind the airfoil. The wind tunnel was operated at three discrete speeds: 20m/s, 35m/s, 50m/s and for each speed interval, the vertical angle of attack was varied, using the VERT rocker switch, through: 0° , 4° , 8° , 12° , and 16° . All this data was collected through VerticalWingVI within the labview system.

Week 3: The airfoil was mounted and connected to strain-gage sting balance to directly measure axial force, normal force, and moment. Before collecting data, zero airspeed (null) readings were taken at each angle of attack to correct any offsets. The tunnel was then operated at 20m/s, 35m/s, and 50m/s and measurements were taken from -4° to 20° in increments of 2° . At each condition, steady state values were recorded, and the null was subtracted to account for offsets and obtain correct aerodynamic loads.

Results

Week 1: By collecting airspeed data with varying wind tunnel power (%) and true airspeed (m/s) versus maximum airflow (%) a linear relationship is determined. This relationship was then used in later parts of the experiment to determine required fan power (%) in order to achieve the required airspeed.

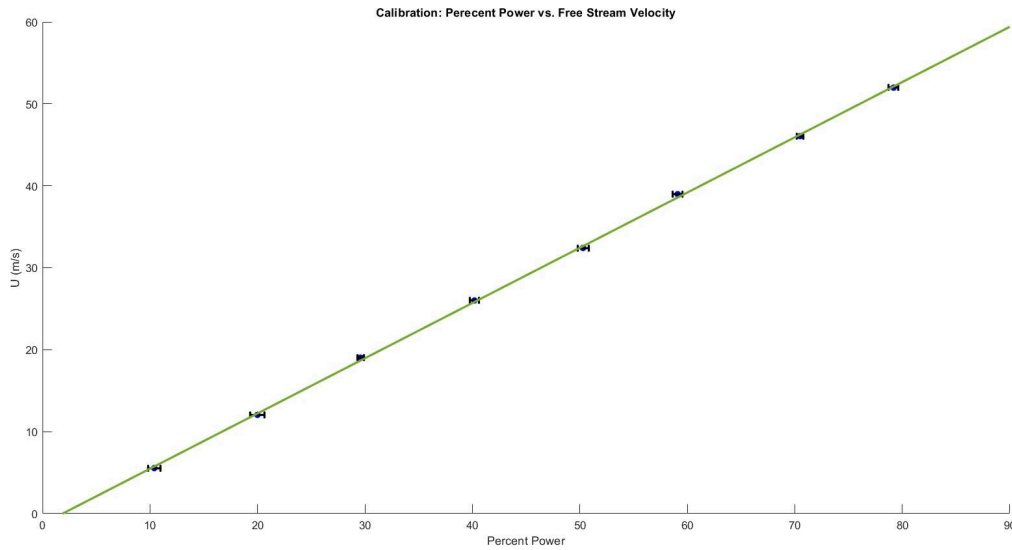


Figure 1: Relationship between percent power input to the wind tunnel and the free stream velocity generated by the tunnel. The data represent a linear relationship adhering to **Equation 16** where $a = 0.674$ and $b = -1.265$. Error bars represent ± 2 standard uncertainty. The normalized

uncertainties for velocity and percent power are $\frac{\widehat{\delta U}_\infty}{U_\infty} \simeq 0.004\%$ and $\frac{\widehat{\delta P}_{perc}}{P_{perc}} \simeq 0.5\%$,

respectively. The uncertainties are smaller than the marker size and are therefore not visually distinguishable on the plot.

Week 2: The raw pressure values were nondimensionalized using dynamic pressure defined in equation (1) and the pressure coefficient relationship given in equation (9). The resulting pressure coefficient distribution was then used in Equation (12a) and (12b) to compute the normal and axial force coefficients through numerical integration along chord. The leading-edge moment coefficients was determined using (13b)

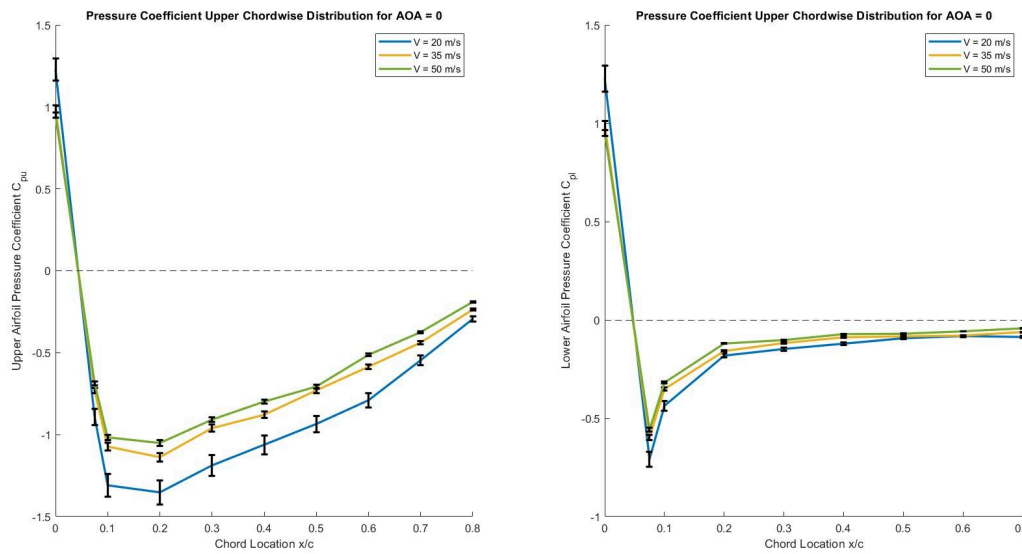


Figure 2: Angle of attack of 0 degrees. Pressure Coefficients plotted against varying chord length (x/c) for velocities of 20 m/s, 35 m/s, and 50 m/s. Error bars represent ± 2 standard uncertainty.

Figure 2 shows the upper surface pressure coefficient distribution at $\alpha = 0^\circ$ for freestream velocities of 20 m/s, 35 m/s, and 50 m/s. The pressure coefficients were calculated using Equation (9) with dynamic pressure defined in Equation (1).

At zero angle of attack, the Clark Y-14 airfoil exhibits an asymmetric pressure distribution due to its cambered shape. A strong peak is observed near the leading edge of the upper surface, while the lower surface experiences a significantly lower pressure drop. The difference between the upper and lower C_p curves indicate a net positive normal force. Based on equation (12a), integrating this pressure difference along the chord results in a positive C_N , confirming the airfoil generates lift at zero angle of attack.

The near overlap of the curves for all three air velocities show that the pressure coefficient distribution is largely independent of freestream velocity when properly nondimensionalized which is consistent with incompressible flow theory

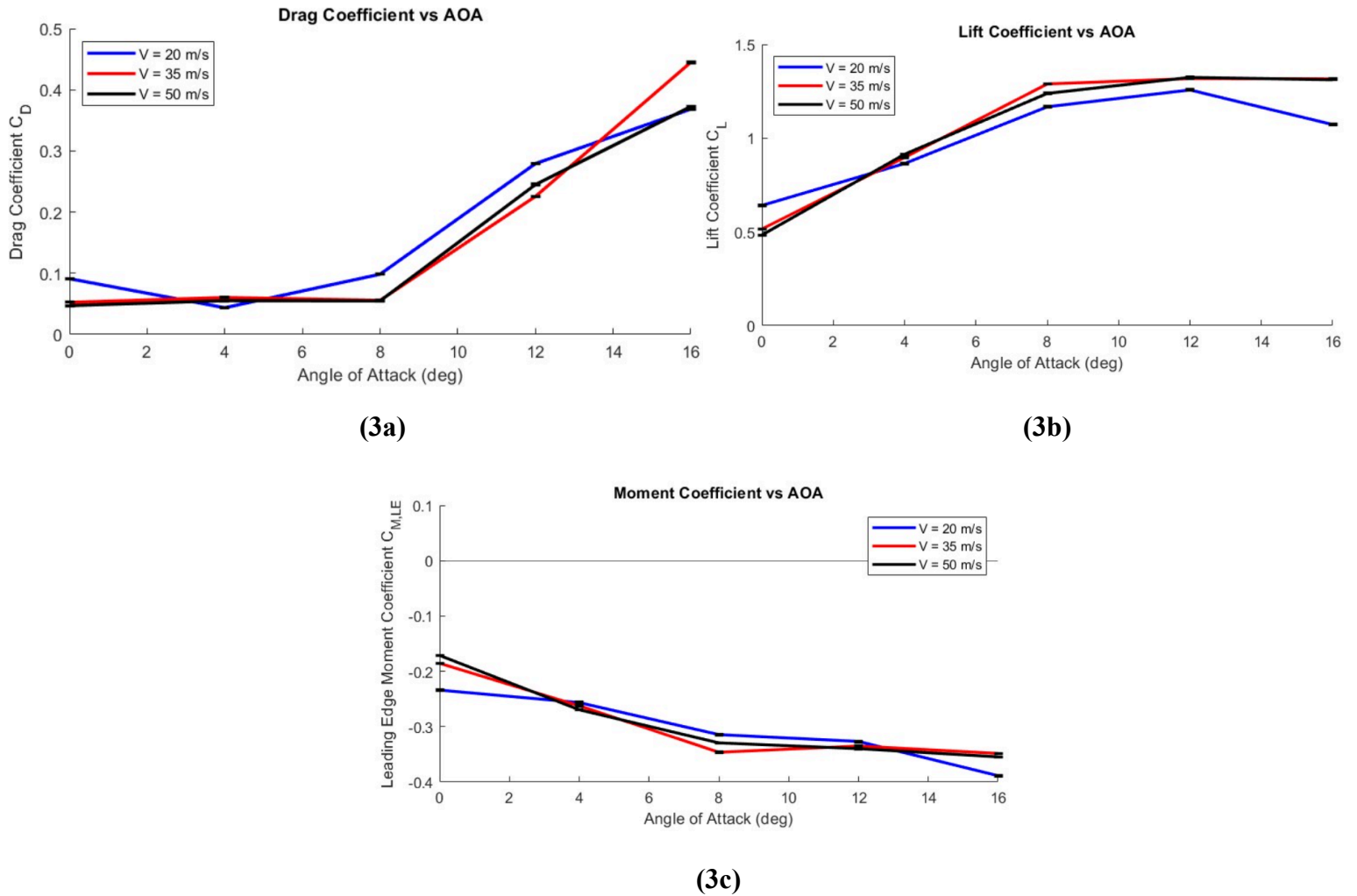


Figure 3: Aerodynamic coefficients plotted against varying angles of attack on a Clark Y-14 airfoil under different velocities. **Figure 3(a):** Moment coefficient plotted against varying angles of attack on a Clark Y-14 airfoil under different velocities **Figure 3(b):** Lift coefficient plotted against varying angles of attack on a Clark Y-14 airfoil under different velocities **Figure 3(c):** Drag Aerodynamic coefficients plotted against varying angles of attack on a Clark Y-14 airfoil under different velocities. Error bars represent ± 2 standard uncertainty.

As Shown in Figure 3(b), at different air velocities, the lift coefficient linearly increases and reaches a peak at $8^\circ - 12^\circ$. The reduction of lift coefficients at higher angle of attack, suggests approaching stall conditions as a result of boundary layer separation over the top

section of the airfoil. The fact that similar angles of attack result in stall for various airspeeds is consistent with theory suggesting lift is highly dependent on the geometry of the airfoil.

The drag coefficients depicted on Figure 3(a) again align with the concept of lift and drag coefficients being independent of airspeed. At the same angles where lift can be observed, steadily decreasing drag in turn increases since boundary layer separation at the top of the airfoil induces drag.

Figure 3(c) depicts the change in moment coefficient with respect to the leading edge. The negative sign indicates a tendency of the nose pitching down. The coefficient again displays behavior largely independent of airspeed as it decreases at similar rates for three different velocities. At the lowest moment coefficient can be located at $\sim 8^\circ$, where the previous plots also indicate stall conditions. This is reasonable as stall conditions result in a negative nose pitch from loss of lift.

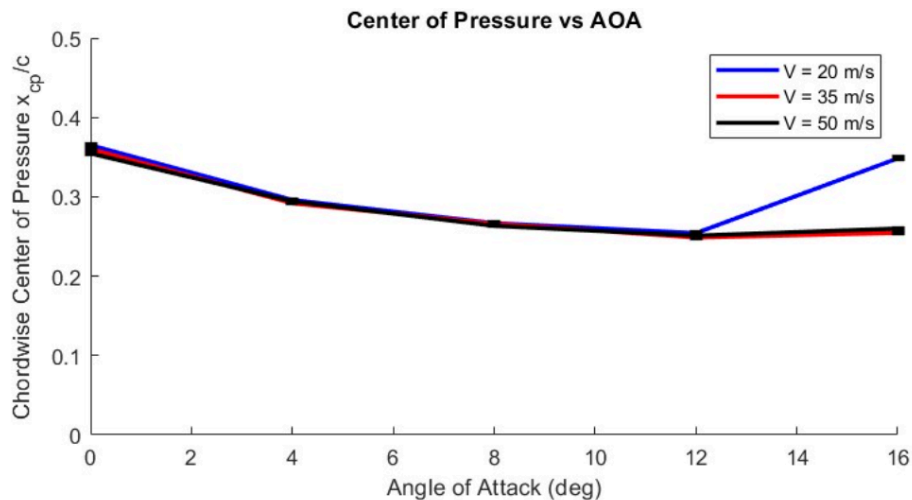


Figure 4: Chordwise Center of pressure plotted against angle of attack on a Clark Y-14 airfoil under different velocities. Error bars represent ± 2 standard uncertainty.

Figure 4 demonstrates the chordwise location of the center of pressure as a function of angle of attack for the three free stream velocities. With an increased angle of attack, the center of pressure tends to move closer to the leading edge.

From 0° to 12° , the center of pressure shifts gradually from approximately $.35c$ to $.25c$. This trend is consistent with increasing lift and the corresponding redistribution of pressure toward the leading edge. The small variation seen between velocities in this section indicates free stream velocity is largely independent of speed under incompressible conditions.

However At 20 m/s there is a noticeable shift in center of pressure at higher angles of attack specifically at 16° . This behavior relates to the reduction in lift at lower velocities and stall conditions approaching, where flow separation alters the pressure distribution and hence shifts the center of pressure rearward.

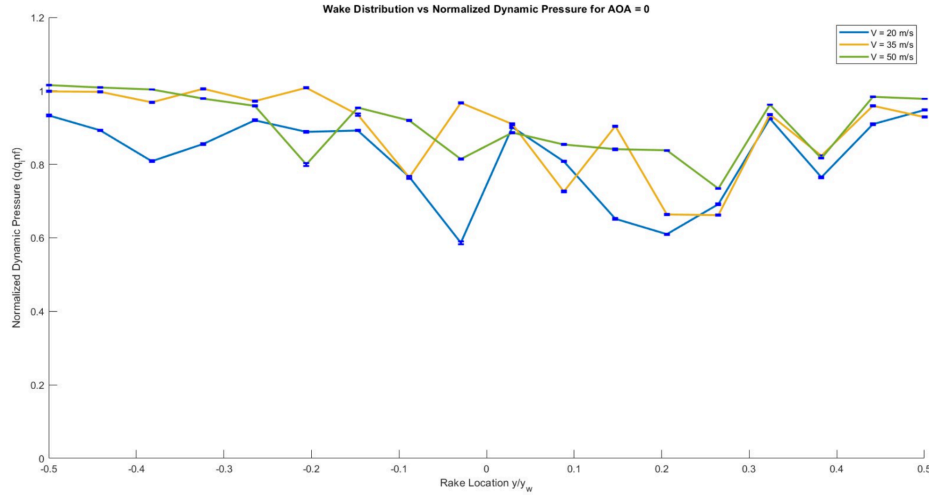


Figure 5: Normalized dynamic pressure distributed across the wake. The x-axis depicts rakewise location normalized by the length of the rake ($y_w = 1.7 \times 10^{-2} m$), and the y-axis depicts the recorded dynamic pressure along the rake normalized by the free-stream dynamic pressure.

Figure 5 shows the normalized dynamic pressure distribution across the wake at an angle of attack at zero at the three freestream velocities. Outside the wake region the normalized dynamic pressure is close to one, indicating nearly freestream conditions. Near the center of the rake, a clear deficit is observed, representing the momentum lost caused by drag on the airfoil.

The 20 m/s case shows a slightly larger deficit, suggesting stronger viscous effects at slower Reynolds number. Overall, the wake profiles confirm the presence of drag and provide the basis for the drag coefficient calculations.

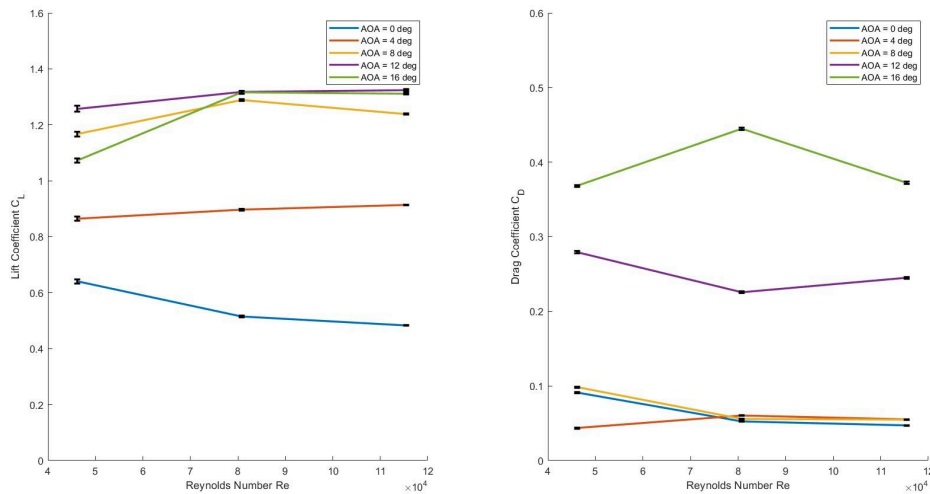
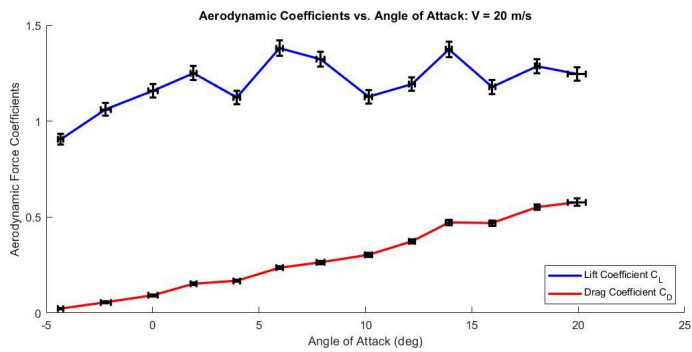


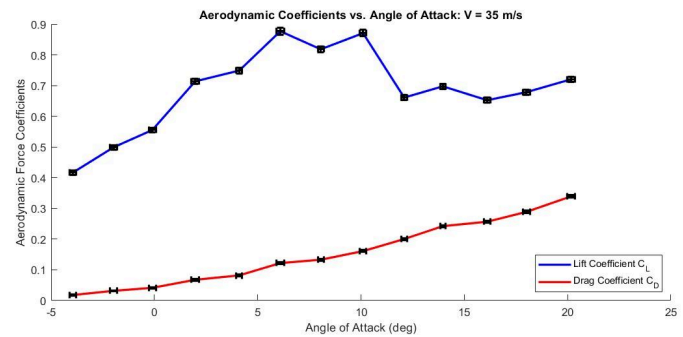
Figure 6: Lift and Drag coefficients calculated at various angles of attack and Reynolds numbers on a Clark Y-14 Airfoil.

Figure 6 illustrates lift and drag coefficients with respect to Reynolds number and differentiates between angles of attack. The lift and drag coefficients stay relatively constant for varying Reynolds numbers. This agrees with the fact that lift and drag coefficients are calculated independently of velocity, as they depend mainly on airfoil geometry, angle of attack, and other aerodynamic variations. The variation of the coefficients with respect to the angle of attacks can be clearly observed as increased angles of attack increase the lift and drag coefficients.

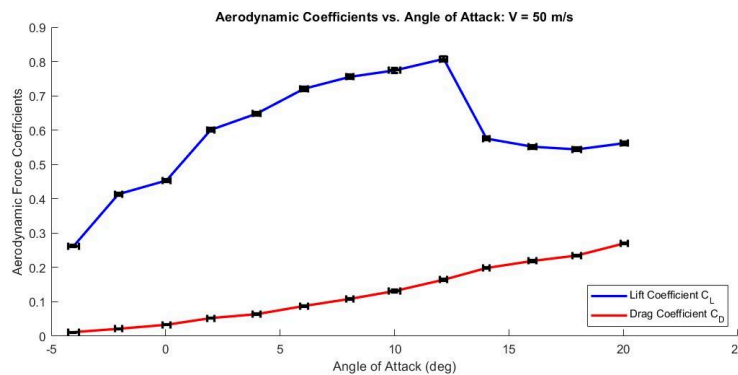
Week 3: Using equations (28) partial derivatives for axial and normal forces as well as moment were calculated and these were converted to lift partial derivatives using equations (29). Then using equations (30), the partial derivatives of lift and drag coefficients were calculated.



(10a)



(10b)



(10c)

Figure 10: Lift coefficient (blue) and Drag coefficient (red) measured for a Clark Y-14 airfoil across three distinct airspeeds: 20m/s, 35m/s, and 50m/s. Lift and Drag coefficients measured against changing angles of attack from -4° to 20° measured in intervals of 2° . Error bars represent ± 2 standard uncertainty.

Interpreting Figure 10, it can be seen that at 20m/s (10a) a maximum lift coefficient occurs at around 6 degrees angle of attack after which stall behavior can be observed with a

decrease in lift. At 35m/s (10b) the maximum lift coefficient can be observed to be at around 10 degrees AOA with stall behavior seen thereafter. At 50m/s (10c) the maximum lift coefficient can be observed at around 12 degrees AOA after which point stall behavior is observed.

As described in these figures above, it can be observed that increasing the airspeed over the airfoils delays the angle of attack at which maximum lift occurs and also delays stall conditions. Ideally, the lift required to maintain steady flight does not need to change, but since the lift and drag coefficients are calculated by normalizing lift and drag with the free stream dynamic pressure, a higher free stream velocity leads to a lower lift and drag coefficient.

Conclusion

This experiment investigated aerodynamic performance of the Clark-Y14 cambered airfoil using different techniques: pressure distribution along and behind the airfoil, wake survey analysis, and sting balance measurement. The general consistency in results between these different methods validated their reliability in experimentally validating expected aerodynamic behavior.

All of the methods used in this experiment demonstrated the linear increase of lift coefficient with angle of attack in pre-stall conditions, which is consistent with thin airfoil theory. Drag coefficients were also observed to gradually increase with angle of attack and more significantly near stall conditions, while the moment coefficient remained relatively constant, likely due to the Clark-Y14's cambered properties. The wake survey method helped to show drag trends, while pressure distribution measurements showed the change in suction peak and pressure recovery with increased angles of attack.

In thin airfoil theory, lift and drag forces increase with airspeed due to the quadratic relationship with dynamic pressure, but for the pressure distribution method, it was observed that aerodynamic coefficients remained largely independent of airspeed. The sting balance measurements showed clear dependence of airspeed and were able to identify the stall regime with nonlinear behavior identified post-stall.

Overall, the experimentally observed results align with expected aerodynamic behavior and help to confirm the validity and reliability of experimental wind tunnel methods in determining lift, drag and moment coefficients of cambered airfoils in low sonic, incompressible regimes.

References

- [1] Seshadri, Kalyanasundaram. MAE 175A: Wind Tunnel Laboratory Handout. University of California, San Diego, Winter 2026. Lab Manual.
- [2] Seshadri, Kalyanasundaram. MAE170, Aerospace Engineering Laboratory Wind Tunnel. University of California, San Diego, Winter 2026. Lecture.

Appendices

Appendix A: Raw Data

Wind tunnel raw data Google Drive link:

https://drive.google.com/drive/folders/1xdjcyb2_-397M0vRBA6Qh9TbqoqFv8Z6?usp=drive_link

Appendix B: Figures

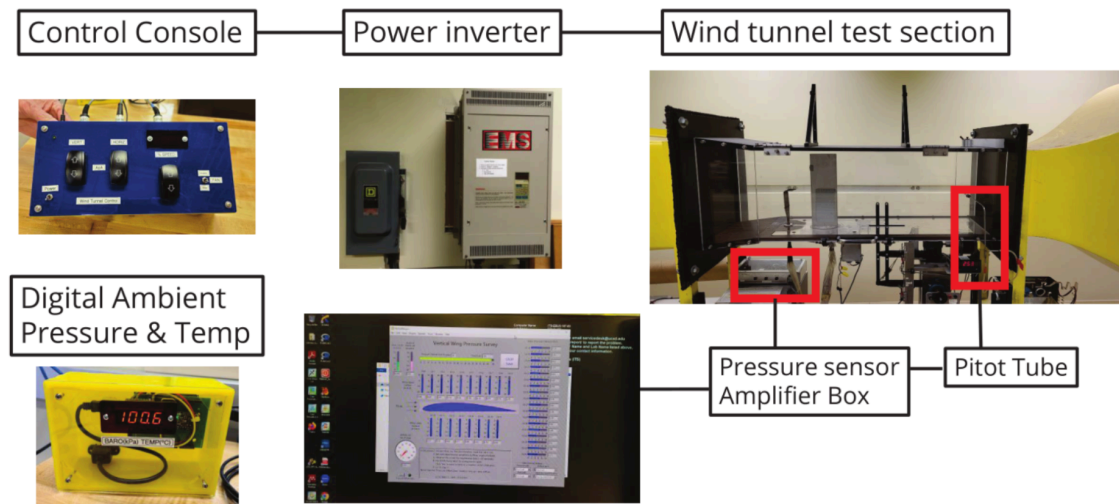


Figure 11 Block diagram of the wind tunnel experimental set up. The test section contains the vertically and horizontally mounted wings along with the pitot tubes. The control console was used to change the angles of attack and increase fan speed. The ambient pressure and temperature sensors were used to determine ambient conditions for calculations.

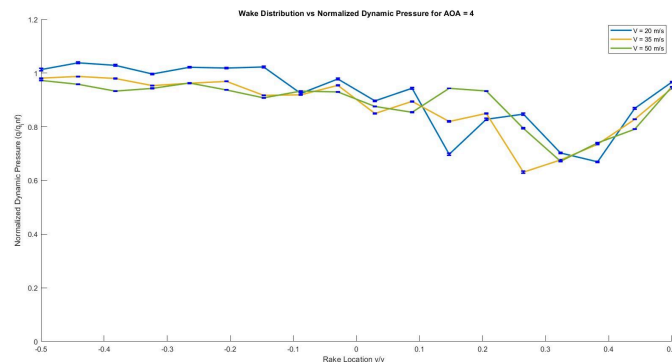


Figure 12a: Normalized wake distribution versus distributed rake locations on a Clark Y-14 airfoil experiencing various velocities at four angle of attack.

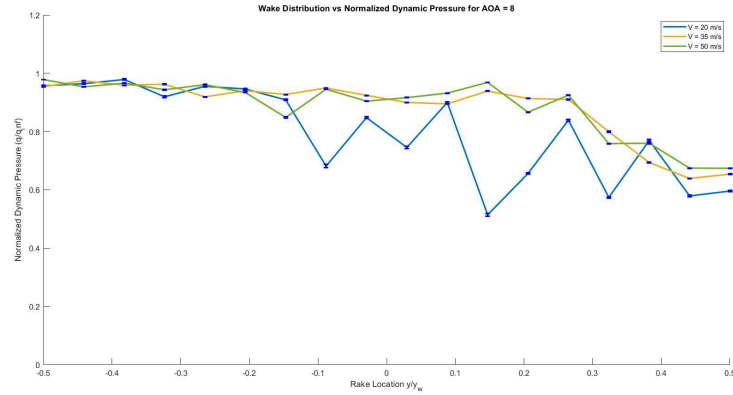


Figure 12b: Normalized wake distribution versus distributed rake locations on a Clark Y-14 airfoil experiencing various velocities at eight angle of attack.

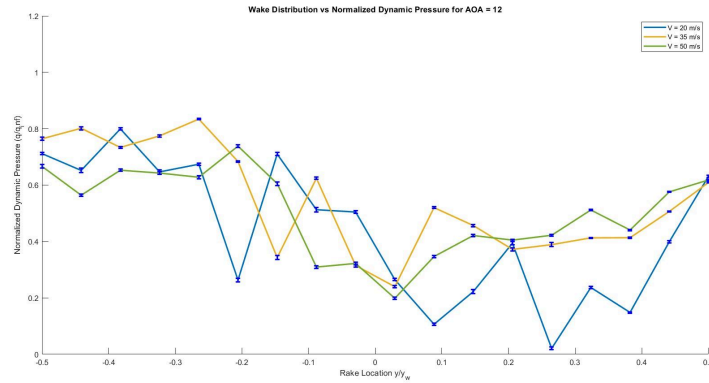


Figure 12c: Normalized wake distribution versus distributed rake locations on a Clark Y-14 airfoil experiencing various velocities at twelve angle of attack.

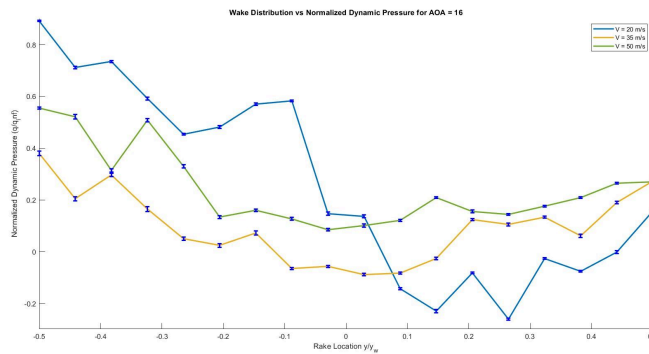


Figure 12d: Normalized wake distribution versus distributed rake locations on a Clark Y-14 airfoil experiencing various velocities at sixteen angle of attack.

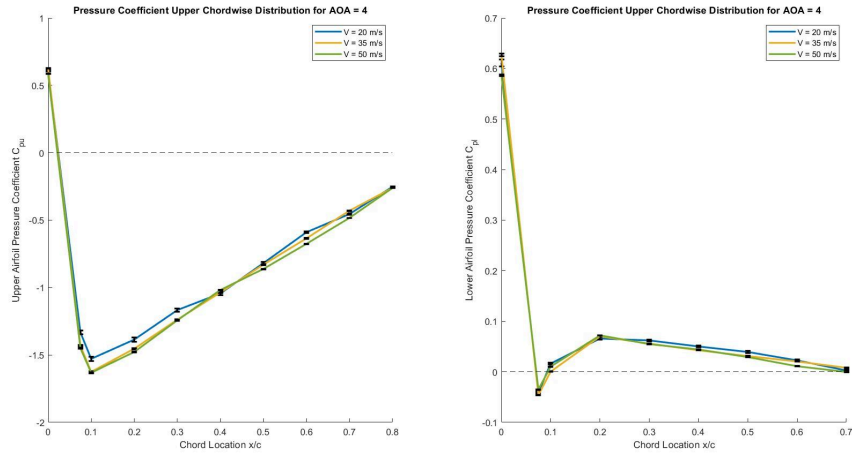


Figure 13a: Angle of attack of 4 degrees. Pressure Coefficients plotted against varying chord length (x/c) for velocities of 20 m/s, 35 m/s, and 50 m/s. Error bars represent ± 1 standard uncertainty. The uncertainties are smaller than the marker size and are therefore not visually distinguishable on the plot.

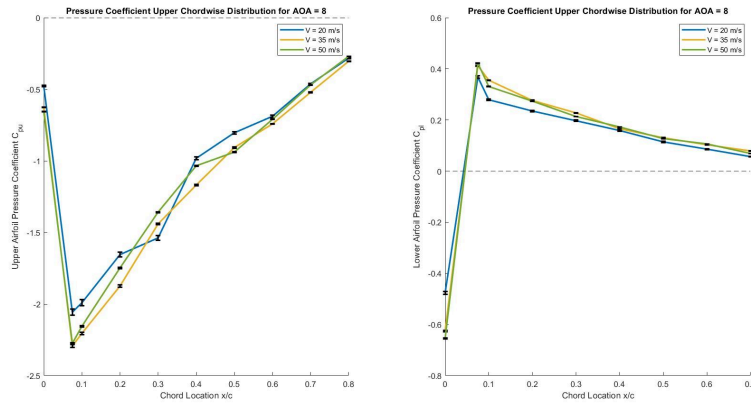


Figure 13b: Angle of attack of 8 degrees. Pressure Coefficients plotted against varying chord length (x/c) for velocities of 20 m/s, 35 m/s, and 50 m/s. Error bars represent ± 1 standard uncertainty. The uncertainties are smaller than the marker size and are therefore not visually distinguishable on the plot.

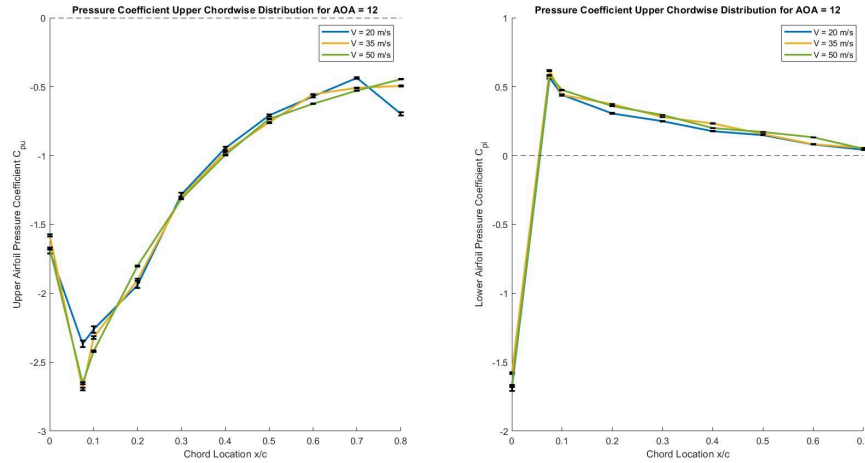


Figure 13c: Angle of attack of 12 degrees. Pressure Coefficients plotted against varying chord length (x/c) for velocities of 20 m/s, 35 m/s, and 50 m/s. Error bars represent ± 1 standard uncertainty. The uncertainties are smaller than the marker size and are therefore not visually distinguishable on the plot.

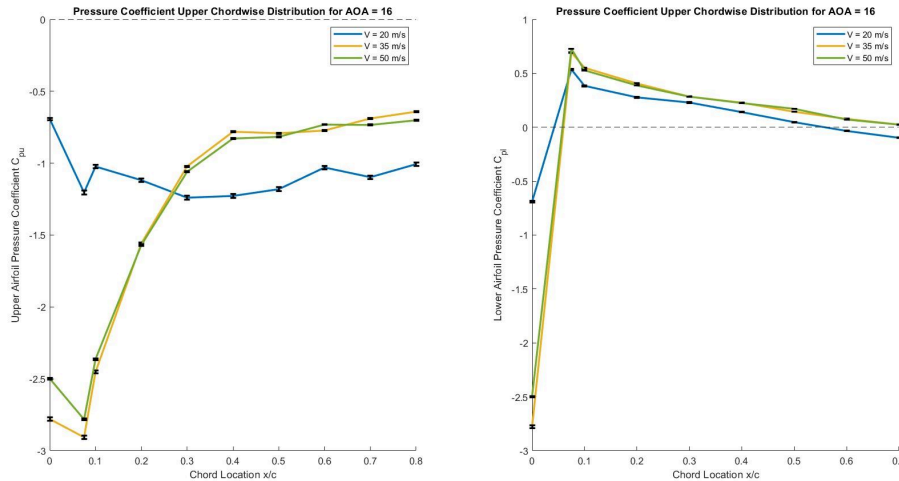


Figure 13d: Angle of attack of 16 degrees. Pressure Coefficients plotted against varying chord length (x/c) for velocities of 20 m/s, 35 m/s, and 50 m/s. Error bars represent ± 1 standard uncertainty. The uncertainties are smaller than the marker size and are therefore not visually distinguishable on the plot.